

APPROVED METHOD:
**LIFE TESTING OF INCANDESCENT
FILAMENT LAMPS**
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-49-20(R2023)

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has been approved by the IES.
Suggestions for revisions
should be directed to IES.

**Prepared by
The IES Testing Procedures Committee**



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CONTENTS

Foreword	1
1.0 Introduction and Scope	1
1.1 Introduction.....	1
1.2 Scope	1
2.0 Normative References	1
2.1 ANSI/IES LS-1-20.....	1
3.0 Nomenclature and Definitions	1
3.1 lamp failure	1
3.2 lamp life.....	1
3.3 life test rack	1
3.4 rated lamp life.....	1
3.5 regulation.....	2
3.6 seasoning	2
4.0 Ambient and Physical Conditions	2
4.1 General	2
4.2 Vibration	2
4.3 Temperature	2
4.4 Airflow	2
4.5 Operating Orientation and Lamp Spacing	2
5.0 Electrical Conditions	2
5.1 Input Voltage	2
5.2 Input Voltage Waveform	2
5.3 Voltage Regulation.....	2
5.4 Electrical Instrumentation.....	3
5.5 Wiring.....	3
5.6 Lamp Holders and Connections	3

6.0 Lamp Test Procedures 3

6.1 Lamp Selection..... 3

6.2 Lamp Handling..... 3

6.3 Lamp Marking..... 4

6.4 Operating Cycle 4

6.5 Recording Failures 4

7.0 Test Report 4

Annex A – Lamp Lumen Maintenance..... 4

Annex B – Determination of the d Exponent..... 5

Informative References 7

Foreword

This approved method is a revision of *IES LM-49-2012, IES Approved Method for Life Testing of Incandescent Filament Lamps*. References were updated.

1.0 Introduction and Scope

1.1 Introduction

This guide describes the procedures by which incandescent lamps can be operated under controlled conditions to obtain optimally comparable data on individual lamp life, changes in light output, and other parameters that vary during the life of the lamp.

Incandescent lamps produce radiant power because of electric current passing through a tungsten filament, which is surrounded by an inert atmosphere or vacuum within a glass envelope. Some lamps contain halogen gases, which are employed to maintain a clean bulb wall. Such lamps may also employ bulb coatings that redirect infrared radiation back to the filament to improve efficacy or to filter visible radiation for color control.

If the filament remains intact, current will flow, heating the filament to incandescence. Since the desired incandescence occurs at high filament temperatures (in excess of 2,000 °C), the surface of the tungsten filament is continually vaporized during lamp operation. As a result, the filament wire diameter is non-uniformly decreased along its length until, at some point, the high current density causes excessive local heating and vaporization, which causes the filament to fail. The rate of evaporation is dependent on the local filament temperature, and gas density and pressure.

It is often important to know the light output, efficacy, and lamp lumen maintenance along with the life of these lamps.¹ **Annex A** discusses this issue.

1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements

during life testing of incandescent filament lamps under standard conditions. This document covers general service incandescent lamps, tungsten-halogen lamps, series airport markers, and street lighting lamps. Not included are automotive, miniature and sub-miniature, photo-optic, and other special lamp types. This Approved Method covers life testing of incandescent lamps at rated voltage or rated current.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Feb 25).

3.0 Nomenclature and Definitions

The units of electrical measurement used in this test method are the volt, the ampere, and the watt.

3.1 lamp failure

For this test method: The point where a lamp stops operating. Other lamp failures, such as manufacturing defects, are reported but not included in the calculation of lamp life.

3.2 lamp life

For this test method: The actual operating time of the lamp until failure, expressed in hours, and shall not include any off time.

3.3 life test rack

A suitable framework or supporting structure where multiple lamp testing positions are available.

3.4 rated lamp life

For this test method: The life value assigned to a particular type of lamp. This is commonly a statistically determined estimate of median operational life. For life rating, the applicable definition of *median* is the

total operating time at which, under normal operating conditions, 50% of any large group of initially installed lamps is expected to be still operating.²

3.5 regulation

The constancy of the voltage or current applied to the equipment under test.

3.6 seasoning

The process of operating a new lamp for a fixed period. This is sometimes referred to as *aging*. A lamp is typically seasoned prior to measuring its initial photometric characteristics. It is an accepted industry procedure for obtaining the “initial” rating of light sources.³

4.0 Ambient and Physical Conditions

4.1 General

It is good laboratory practice that the storage and testing of lamps be undertaken in a relatively clean environment. Because lamps operate at high bulb temperatures, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before mounting on the life test racks.

4.2 Vibration

Lamps should not be subjected to excessive vibration or shock during life testing.

4.3 Temperature

Ambient temperature is not a critical consideration for the life testing of incandescent lamps. Bulb wall temperature and seal temperature for tungsten-halogen lamps should remain within the specifications of the manufacturer.

Care should be taken not to exceed the specified operating temperature range for the equipment and instrumentation.

4.4 Airflow

Airflow will not affect the life of incandescent lamps but will aid in cooling the life test area and the life test rack electrical components.

4.5 Operating Orientation and Lamp Spacing

The requestor of the test should specify the operating orientation of the lamps to be tested. Where an orientation is not specified, or where more than one orientation is specified, the lamp should be tested in the orientation that will be used in the application.

The lamps shall be spaced to allow airflow around each lamp. This is facilitated by designing open life-testing racks with minimal structural components to block airflow.

5.0 Electrical Conditions

5.1 Input Voltage

Alternating current (AC) operation is the standard for incandescent lamp life testing. Direct current (DC) operation may be used to emulate specific application conditions.

Lamps shall be operated at their rated voltage or current for a current-rated lamp, and at 60 Hz. Input voltage shall conform to the rated root-mean-square (RMS) input voltage and frequency of the auxiliary equipment. If DC or another frequency is used, it shall be noted on the test report. If the rated voltage or current is a range, the center value shall be used and reported as a test condition.

5.2 Input Voltage Waveform

For AC power operation, the power supply shall have a sinusoidal voltage waveform such that total harmonic distortion (THD) does not exceed 3% of the fundamental frequency. Note: THD shall be calculated as the root-mean-square (RMS) summation of the harmonic components (orders of magnitude of 2 to 50 for a 100-kHz meter, and orders of magnitude of 2 to 100 for a 1-MHz meter, as a minimum) divided by the fundamental frequency during operation of the DUT.

5.3 Voltage Regulation

The AC RMS voltage, or RMS current, shall be monitored and regulated to within $\pm 0.5\%$ of the rated RMS value for the duration of the life test. If operated on DC, the

voltage (or current) shall be monitored and regulated to within $\pm 0.5\%$ of the rated value for the duration of the life test.

Variation in the input voltage during the life test can have a significant impact on the measured lifetime of the test lamp. Incandescent lamp life is known to be proportional to about the 13th power of the voltage. Hence, an average change of 0.1% in input voltage over the duration of the test will lead to a change in lamp life of over 1%. It is recommended that the life test system be designed to be controlled to within 0.1% if possible.

5.4 Electrical Instrumentation

Accurate recording of elapsed operating time is important in lamp life testing. If used, an elapsed-time recording device shall accumulate time only when the associated lamps are operating. Video monitoring, current monitoring, or other means may be used to determine elapsed operating time if designed to provide sufficient temporal accuracy. Additional guidelines on instrumentation are available.^{2,4}

To facilitate the periodic checking of auxiliary equipment or lamp characteristics, appropriate test receptacles may be provided to permit instruments to be connected to each circuit. In AC installations, instruments shall measure true root-mean-square (RMS) values.

Electrical instrument manuals provide values for the tolerance (accuracy) of the instrument, normally as a percentage of the reading. The instrument tolerance of the measurement of voltage for the life test shall be:

- For RMS AC voltage, $\pm 0.5\%$ or less
- For DC voltage, $\pm 0.05\%$ or less

5.5 Wiring

All wiring of life test racks shall be in accordance with national, state or provincial, and local electrical codes. Racks should be wired such that the voltage drop across any number of lamp holders wired in parallel does not exceed 0.1%.

Lamps that require series circuits, such as series airport markers and street lighting lamps, shall be equipped with cutout receptacles or transformers so that when a lamp in the test fails, the remaining lamps will continue to burn.

5.6 Lamp Holders and Connections

For screw-base lamps, lamp holders should be of a porcelain-type construction with sturdy threaded shells and center contacts. Lamp holders shall be rated for the specific test application under which they are used. Nickel or silver-plated shells and center contacts are preferred over plain brass because of their resistance to corrosion. Aluminum is not recommended.

For non-screw-base lamps, the same principles of thermal and mechanical stability shall be applied to the lamp mounting mechanism.

Inspection of lamp holder contacts and verification of the input voltage should be made whenever lamps are placed on the life test rack. Lamp holders are to be replaced if they show signs of weak center contacts, corroded shells, or worn insulation.

Additional information on lamp holder and connection standards is available.^{5,6}

6.0 Lamp Test Procedures

6.1 Lamp Selection

The lamps shall be selected to represent the purpose of the test. However, if the test is designed to evaluate specific characteristics, the sample selection needs to take this into consideration. The value of the test will depend upon the method of sampling, size of the sample, conditions of testing, and other factors.

6.2 Lamp Handling

Prior to operation, lamps shall be cleaned to eliminate surface contamination.

When lamps are installed or removed from life test positions, the voltage to the lamp holder shall be turned off. Installing lamps with the electrical power turned on will cause shock to the filament and could severely reduce lamp life.

If the lamps are to be removed from the life test positions for photometric testing, the lamps are to be

handled, transported, and stored in such a way as to minimize the effect of vibration and shock. The lamps should cool to ambient temperature prior to being disturbed.

Bare quartz capsules should not be handled with bare hands, as human skin oils can cause devitrification and failure of the quartz.

6.3 Lamp Marking

Individual lamps shall be tracked during life testing. Lamps can be identified by markings applied directly to the lamps, or by labels that can be attached to the lamps during transport and evaluation, or to the life test rack position occupied by the lamps during a life test. The identification method selected should consider the effect of exposure to light, UV and IR radiation, and heat.

Possible suitable marking methods and materials include durable bar coding, ceramic ink marking, high temperature markers, and any other method which endures or can be periodically renewed for the length of the life testing.

6.4 Operating Cycle

Life test lamps should be allowed to cool to ambient temperature once per day. This cooling period is usually 15 to 30 minutes per day. The cycle shall be noted in the test report.

Accelerated life testing is possible with non-tungsten-halogen incandescent lamps. Operating the lamps at higher than rated voltage will cause the lamp to fail faster than at rated voltage. A scaling factor is then applied to the measured lifetime to determine the predicted lifetime. This scaling factor should be determined from parallel or previous life testing at rated voltage.

6.5 Recording Failures

Checking for lamp failures either by visual observation or automatic monitoring shall be done at intervals of no more than 1% of rated life. The recorded failure time shall be the midpoint of the last monitoring interval. Each failure shall be investigated to make certain that it is a lamp failure and is not caused by improper functioning of the auxiliary equipment or lamp holders.

Lamp failures can be divided roughly into three classes: those caused by manufacturing defects, those caused by normal failures, and lamps showing markedly low light output.

7.0 Test Report

Test reports shall include the hours at which lamps failed for each lamp installed at the start of the test.

The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamp types being tested. Typical items reported are:

- Number of lamps tested
- Selection criteria
- Description of lamp type
- Operating cycle
- Ambient conditions (refer to **Section 4**)
- Test voltage or current
- Lamp orientation during operation
- Number of elapsed hours at which lamp failed
- Class of failure of each lamp
- Lamp monitoring interval
- Median life of entire group
- Special test conditions (Example: minimum light output of an individual lamp type before it is considered to have failed; any unusual conditions observed.)

Annex A – Lamp Lumen Maintenance

It is often desirable to measure the lamp lumen maintenance during a life test. This function of lamp performance is determined from measurement of the photometric characteristics of the lamp during the life test, typically expressed as a percentage of rated output. Photometric measurements are made after at least two intervals, customarily 40% and 70% of rated life. These values are compared to the initial lamp

output measured after the initial seasoning time of the lamp (see **Section 3 Nomenclature and Definitions**). Additional intervals can be used to further define the shape of the lamp lumen maintenance curve.

Annex B – Determination of the *d* Exponent

B.1 Definition

For constant voltage operation of incandescent lamps, the relationship between the lamp life and applied voltage can be expressed as:

$$\frac{Life}{LIFE} = \left(\frac{VOLTS}{Volts} \right)^d, \quad (B-1)$$

where:

- *d* is the life-voltage exponent for certain incandescent lamps when they are operated under constant-voltage conditions; this is the value to be calculated and, once determined, will enable determination of the rated life of the lamp
- The lowercase quantities are the variables and the uppercase quantities are the rated values

By using natural logarithms, **Equation B-1** can be converted so the *d* exponent becomes expressed as:

$$d = \frac{-d \ln(Life)}{d \ln(Volts)} \quad (B-2)$$

Note: Each of the non-italicized *d*'s on the right-hand side of **Equation B-2** denote a differential operation in mathematics and have no relation to the italicized *d*, the value to be calculated.

B.2 Experimental Determination

Experimental values for the *d* exponent for a given lamp type can be obtained by life testing groups of lamps at various fixed voltages. The slope of the tangent to the $\ln(Life) - \ln(Volts)$ curve yields the value of *d* at any given point. In principle, the *d* exponent changes with voltage for a given lamp type, but in practice the change in the value of *d* is small enough over a range of force voltages.

Since the variation in *d* with voltage is small and the uncertainty in life is large, the use of the *d* exponent has

been found to be useful for determining rated life of new lamp designs via accelerated life testing. The use of accelerated life testing for tungsten-halogen lamps is usually discouraged, as it is often found that the tungsten atom transport "rate" mechanism (see **Section B.3 Theoretical Basis**) changes with applied voltage. If the tungsten transport rate mechanism is known not to change, then accelerated life testing for tungsten-halogen lamps can be used. This often means that the accelerating voltage cannot be raised as much as it would be for tungsten filament lamps; i.e., determining tungsten-halogen lamp life from accelerated testing will take longer.

B.3 Theoretical Basis

The assumption is made that lamp life is inversely proportional to a rate process that involves the transport of tungsten atoms. This rate mechanism is usually found to be due to evaporation, but for very fine wire or for very long lifetimes the rate limiting mechanism has been found to be due to surface diffusion. The discussion here is limited to evaporative rate processes. Thus:

$$Life \propto \frac{1}{Rate} \quad (B-3)$$

The development below follows that published by E.J. Covington in the Journal of the IES in 1973.⁷ Other references can be obtained from that publication.

Taking the natural log of **Equation B-3** and substituting into **Equation B-2** results in:

$$d = \frac{d \ln(Rate)}{d \ln(V)} = \frac{d \ln(Rate)}{d \ln(T)} \cdot \frac{d \ln(T)}{d \ln(V)}, \quad (B-4)$$

where:

T is the temperature in kelvins

V is the same as *Volts* in **Equation B-2**

The evaporation rate of tungsten is known to follow the Arrhenius equation:

$$Evaporation\ rate = C \exp(-Q/RT), \quad (B-5)$$

where:

Q is the activation energy of the process

R is the gas constant

Since R is a constant, the exponent can be written as:

$$-H/T$$

Substituting **Equation B-5** into **Equation B-4** gives:

$$d = \left(\frac{H}{T}\right) \left(\frac{d \ln(T)}{d \ln(V)}\right) \quad (\text{B-6})$$

Measurements of the evaporation rate of tungsten by several people yield values for H between 100,000 and 110,000. A value of 105,000 is used in the example below.

The next step is to determine the ratio of the rate of change of temperature with voltage, i.e., the far-right part of **Equation B-6**. This can be found from the known temperature relationships of resistivity and radiated power of tungsten, as expressed by the energy balance for the wire. Thus:

$$V^2 = R \cdot W \quad (\text{B-7})$$

For constant wire diameter and wire length (same filament), **Equation B-7** can be formulated in terms of the temperature:

$$(V/V_0)^2 = (T/T_0)^{1.2} \cdot (T/T_0)^n \quad (\text{B-8})$$

Here, the 0 subscripts denote reference or rated values. The two terms on the right-hand side of **Equation B-8** represent the temperature dependence of the resistance and output power (dominated by the radiative process), respectively. Taking the natural log and differentiating **Equation B-8** leads to:

$$\left(\frac{d \ln(T)}{d \ln(V)}\right) = \frac{2}{1.2 + n} \quad (\text{B-9})$$

The resistivity of tungsten is known to vary as the 1.2-power of the temperature, independent of other factors. The issue is the value of n , the exponent for the output power. Since tungsten is nearly a blackbody, the radiated power varies nearly as the fourth power of the temperature, where the variation is due to the total emissivity of tungsten, which is known to increase with temperature with a varying exponential value. For temperatures around 2500 K to 3000 K, the exponent for the radiated emissivity is about 4.6, decreasing slowly as the temperature increases. Thus, for vacuum lamps, where radiative losses account for almost all the output power, the value of n is about 4.6 when the filament temperature is on the order of 2500 K.

For gas-filled lamps, some of the output power goes into heating the gas. The exponent for the temperature dependence of the gas loss is about 1.7, much lower than the exponent needed to account for radiation. Thus, for gas-filled lamps the combined exponent due to both radiated and gas-loss power depends on the temperature, fill gas type, pressure, and filament design. For most A-line CC8 designs, the combined value for n is around 4.0. On the one hand, the higher the gas-loss fraction, the lower the value of n , while on the other hand, the higher the temperature, the lower the value of n . Thus, increasing filament temperature, as in accelerated testing, tends not to change the value of n significantly.

Finally, combining **Equation B-6** and **Equation B-9**:

$$d = \left(\frac{H}{T_a}\right) \left(\frac{2}{1.2 + n}\right) \quad (\text{B-10})$$

It should be noted that the d exponent is inversely proportional to the temperature, which limits the range over which the value of d is valid. Further, since it is necessary to account for this linear dependence with T , it is important to choose the temperature correctly. It is recommended that the temperature of the filament at the accelerated voltage and the temperature of the filament at rated voltage be averaged; this is denoted by T_a in **Equation B-10**.

B.4 Example

Given: an incandescent lamp filament rated at 120 volts. At rated voltage this lamp filament operates at a wire temperature of 2800 K. It is assumed to be a gas-filled lamp and that $n = 4.0$.

It is desired to accelerate testing at 150 volts. What is the d exponent?

To determine the d exponent it is first necessary to determine the temperature of the filament when operated at 150 volts. From **Equation B-8**:

$$T = T_0 \left(\frac{V}{V_0}\right)^{0.385} \quad (\text{B-11})$$

Inserting the numbers into **Equation B-11** yields a filament temperature of 3051 K. Thus, the average temperature between 120 and 150 volts is: $(2800 + 3051)/2 = 2925.5$ K.

This is the value of T_a needed to insert into **Equation B-10**. Inserting the known values of T_a , H , and n into **Equation B-10** yields:

$$d = \left(\frac{105,000}{2,2925.5} \right) \left(\frac{2}{5.2} \right) = 13.8$$

REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LM-45-20, IES Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: IES; 2020.
- 2 Illuminating Engineering Society. ANSI/IES LP-4-20, Lighting Practice: Electric Light Sources – Properties, Selection, and Specification. New York: IES; 2020.
- 3 Illuminating Engineering Society. ANSI/IES LM-54-20, Approved Method: IES Guide to Lamp Seasoning. New York: IES; 2020.
- 4 Illuminating Engineering Society. ANSI/IES LM-28-20, Approved Method: IES Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.
- 5 National Electrical Manufacturers Association. ANSI C81.62-2017, American National Standard for Electric Lampholders. Washington, DC: American National Standards Institute; 2017.
- 6 UL, LLC. ANSI/UL 496-2017, Lampholders, American National Safety Standard. Northbrook, Ill.: UL, LLC; 2017.
- 7 Covington EJ. The life-voltage exponent for tungsten lamps. J Illum Engineering Soc. 1973;2(2):83.



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